

Curriculum Vitae

Imran Khan

Adjunct Lecturer and Practitioner-in-Residence

Contact Information

City University of New York - Hunter College
School of Arts and Sciences, Department of Computer Science
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Teaching Interests

Artificial intelligence, machine learning, programming, scientific models, statistics

Education

Rice University 2020-2023
Jones Graduate School of Business
Master of Business Administration (MBA)
Capstone: Play Forever (Canada)

New York University 2017-2019
Tandon School of Engineering, Center for Urban Science and Progress
Master of Science (MS) - Applied Urban Science and Informatics
Capstone: Quantifying segregation in New York City schools (with machine learning clustering)

University of Virginia 2011-2013
Batten School of Leadership and Public Policy
Master of Public Policy (MPP)
Capstone: Green procurement strategic plan: Policy analysis, planning, and implementation of sustainable New York City operations

Southern Methodist University 2009-2010
Simmons School of Education and Human Development, Department of Teaching and Learning
Master of Education (MEd) w/ Honors
Thesis: Evaluation and implications of drop-out prevention program among minority adolescents

Texas A&M University 2000-2003
College of Agriculture and Life Sciences, Department of Biochemistry and Biophysics
Bachelor of Science (BS) w/ Honors - Biochemistry
Thesis: Protein peptide purification of Human Metal-responsive Transcription Factor 1

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Teaching Experience

City University of New York - Hunter College 2022-Present
Course taught: Relational Databases and SQL Programming

The Carpentries 2022-Present
Courses taught: Data Carpentry, Library Carpentry, Software Carpentry

City University of New York - Energy Management Institute 2020-Present
Course taught: Problem Solving Using Microsoft Excel

City University of New York - LaGuardia Community College 2020-Present
Courses taught: Data Analytics Foundations, Data, Databases and SQL, Data Visualization with Tableau, Power BI for Data Analytics, Predictive Analytics: Machine Learning, Python for Data Analytics

Community College of Philadelphia 2024-2025
Courses taught: Database Management Systems, Foundations of Data Science, Introduction to Information Technology, Machine Learning

Garland Independent School District 2010-2011
Course taught: Chemistry

Collin College 2010
Course taught: Chemistry

Pegasus School of Liberal Arts and Sciences 2009
Course taught: Chemistry

Pasadena City College 2007
Course taught: Chemistry

California State University, Los Angeles 2004-2006
Course taught: Chemistry

Research Experience

Texas A&M University 2025-Present
High Performance Research Computing

New York University 2019
Urban Modeling Group

New York University 2018
GovLab

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Southern Methodist University 2010-2011
John Goodwin Tower Center for Political Studies Student Forum

Texas A&M University 2003
United States Department of Agriculture, Agricultural Research Service

Texas A&M University 2001-2004
College of Agriculture and Life Sciences, Department of Biochemistry and Biophysics

Industry Experience

CivicReach 2026-Present
Government Partnerships Broker

Hudson Shipping Corporation 2025-2026
Co-Founder and Chief Operating Officer

LaGuardia APEX 2024
Procurement Specialist

Cyber Agility Academy 2024-2025
Co-Founder and Chief Operating Officer

Ernst and Young Services Ltd. 2022-2024
Engagement Manager (Financial Services, Extended Assurance, Internal Audit, Digital Risk and Innovation)

Atkinson-Noland and Associates 2020
Engineer
Certification: OSHA Construction (30-Hr)

Mumble Solutions 2019
Software Engineer - Lead Developer

Arcadis 2018
Data Scientist

Build Well Labs 2018-2026
Principal
Clients: Datapolitan, Education Pioneers, Egerton Clinic, Healthfirst, Movement Labs, New York Academy of Sciences, New York City Department of Education, Philosophy LLC, United States Department of Defense, United States Department of Health and Human Services
Pro Bono Clients: American Chemical Society, Code for America, Company OnStage, Cooking for Community, DataKind, Dusit International, Mandela Washington Fellowship for Young African Leaders, New Politics Leadership Academy

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Acumen Solutions Consultant	2013
Chemical Information Services Chemical Data Researcher	2009-2010
Teva Pharmaceuticals (formerly Watson Pharmaceuticals) Scientist Certification: CGMP (Current Good Manufacturing Practices)	2008
3M Materials Scientist	2007
Curbside, Inc. Chemist and Hazardous Materials Specialist Certification: OSHA HAZWOPER (40-Hr)	2007

Government Experience

New York City Mayor's Office Office of Operations, Office for Economic Opportunity Director of Special Projects Assignments: Education, Housing Authority, Service, Social Services, Young Men's Initiative, Youth and Community Development	2016-2017
Federal Bureau of Investigation Citizens Academy	2015
New York Police Department Citizens Police Academy	2015
New York State Governor's Office Department of Agriculture and Markets, Division of Agricultural Development Empire State Fellow Assignments: Anti-Hunger Task Force, Community, Opportunity, Reinvestment Neighborhoods Initiative, Lean Spending and Government Efficiency Commission, Long Island, Mid-Hudson, and New York City Regional Economic Development Councils, Office of General Services, Rochester Anti-Poverty Task Force	2013-2015

Fellowships

United States Army DEVCOM Army Research Laboratory Foundry Program via FedTech, National Security Innovation Network	2022
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Rice University 2021-2022
Fondren Library
Fondren Fellowship
Certification: Software Carpentry Instructor

United States Department of Commerce 2020
Census Bureau
Civic Digital Fellowship via Coding it Forward

University of Southern California 2018
Dornsife College of Letters, Arts and Sciences, Center for Religion and Civic Culture
American Muslim Civic Leadership Institute Fellowship

New Leaders Council 2014
Capital District New York
Fellowship

State University of New York 2013-2014
Rockefeller Institute of Government
Empire State Fellowship

New York City Department of Citywide Administrative Services 2013
Office of Citywide Procurement
Green Procurement Fellowship

United States Department of Energy 2012
Office of Economic Impact and Diversity
Minority Educational Institution Student Partnership Program Fellowship

Houston Independent School District 2011
School Improvement Office
Education Pioneers Summer Graduate School Fellowship

California Institute of Technology 2004-2007
Division of Chemistry and Chemical Engineering
National Science Foundation Partnership for Research and Education in Materials Fellowship

Service Projects

Delaware Division of Libraries 2024-2025
Dover, DE

BakerRipley 2020-2021
Houston, TX

New York City Mayor's Office of Criminal Justice 2018

Curriculum Vitae

New York, NY

New York City Department of Education 2015
New York, NY

CitySquare 2010
Dallas, TX

Board Appointments

Service Year Alliance 2025-Present
Alums Amplified Speakers Bureau Member

University of Pennsylvania, College of Arts and Sciences 2025-Present
South Asia Center Affiliated Faculty Member

Harvard Business Review 2024-Present
Advisory Council Member

LaGuardia Community College 2026
Artificial Intelligence Certification Program AI Advisory Board Member

LaGuardia Community College 2025-2026
Generative AI Working Group Member

LaGuardia Community College 2024-2025
AI Advisory Board Member

American Bar Association 2024-2024
Model Procurement Code Revision Drafting Committee Member

techUK 2023-2024
TechSkills Data Science Working Group Member

Cyber Readiness Institute 2023-2024
Small Business Advisory Council Member

American Chemical Society, New York Local Section 2019-2020
Computers in Chemistry Topical Group Co-Chair

New York City Human Resources Administration 2015
SNAP Task Force Member

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Bedford Stuyvesant Restoration Corporation Central Brooklyn Food Hub Task Force Member	2015
Cornell Small Farms Program Northeast Sustainable Agriculture Research and Education Advisory Board Member	2015
American Farmland Trust Farm to Institution New York State Leadership Team Member	2014-2015
New York State Energy Research and Development Authority Technical Evaluation Panel Member	2014

Current Professional Association Membership

American Chemical Society

Association for Supply Chain Management

Engineers' Club of Philadelphia

Research Data Alliance

United States Digital Response

United States Research Software Engineer Association

Publications and Reports

Bateman, A., Grove, L., Swartz, H., and Yang, Z., and Khan, I. "Linking Long Island agriculture to New York City public schools," University of Virginia Frank Batten School of Leadership and Public Policy, 2015.

Dudley, R., Goodman, D., Hunt, J., and Jung, J., and Khan, I. "Growing local food systems: Policies to expand market access for New York's small farmers," University of Virginia Frank Batten School of Leadership and Public Policy, 2014.

Connors, K., Khan, I., Orphanides, E., and West, M. "Income-generating programs: Best practices and recommendations for Lebanon," University of Virginia Frank Batten School of Leadership and Public Policy, towards partial fulfilment of Master of Public Policy coursework and degree requirements, 2012.

Curriculum Vitae

Mathias, E.V., Liu, X., Franco, O., Khan, I., Ba, Y., and Kornfield, J.A. “A model of drug-loaded fluorocarbon-based micelles studied by electron-spin induced ^{19}F relaxation NMR and molecular dynamics modeling,” *Langmuir*, 24, 692-700, 2008.

Conference Invitations and Talks

Vermont Biomedical Research Network, Software Carpentry, June 2026, Burlington, VT

Texas A&M University, Division of Research, High Performance Research Computing, Building Research Innovation at Community Colleges Research Data Management Conference 2026, March 2026, Alexandria, VA

University of Pennsylvania, Data Carpentry, March 2026, Philadelphia, PA

University of Massachusetts Dartmouth School for Marine Science and Technology, Software Carpentry, October 2025, Dartmouth, MA

Texas A&M University, Division of Research, High Performance Research Computing, Summer PACES Research Training Workshop, August 2025, Albuquerque, NM

Texas A&M University, Division of Research, High Performance Research Computing, Building Research Innovation at Community Colleges Research Data Management Conference 2025, July 2025, Alexandria, VA

University of Massachusetts Dartmouth School for Marine Science and Technology, Data Carpentry, November 2024, Dartmouth, MA

University of Hawaii, Data Carpentry, October 2024, Honolulu, HI

University of Pennsylvania, Data Carpentry, June 2024, Philadelphia, PA

Pennsylvania State University, Software Carpentry, April 2024, State College, PA

Michigan Technological University, Data Carpentry, February 2024, Houghton, MI

University of Sussex, Data Carpentry, July 2023, Brighton, England

University of Pennsylvania, Penn Graduate Consulting Club, November 2022, Online

University of Pennsylvania - Van Pelt Library, Data Carpentry, October 2022, Philadelphia, PA

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The i4 Group, Data-Driven Service Development, June 2022, Online

The i4 Group, Using Open Data for Public Benefit, March 2022, Online

The i4 Group, Data Driven Product Development, August 2021, Online

General Assembly, Data Week | Let's Break it Down: Data Science vs. Data Analytics, May 2021, Online

Long Island Food Conference, April 2015, Hempstead, NY

Farm to Institution Summit, April 2015, Amherst, MA

AcademyHealth, National Health Policy Conference, February 2015, Washington, DC

Cornell University, Cornell Cooperative Extension, New York Farms to New York Plates: The Becker Forum on Institutional Purchasing Across NYS, January 2015, Syracuse, NY

American Farmland Trust, New York State Office, Hudson Valley Farm to Institution Summit, October 2014, New Paltz, NY

Heritage Radio Network, The Future of Food Leadership: Feeding NYS, September 2014, Brooklyn, NY

Empire Farm Days, August 2014, Seneca Falls, NY

White House, Summit on Working Families, June 2014, Washington, DC

New York State Office of General Services, Purchasing Forum and Trade Show, May 2014, Albany, NY

SOMOS El Futuro, Spring Conference, March 2014, Albany, NY

New Leaders Council, Northeast Conversation, March 2014, Jersey City, NJ

Asian American Government Executives Network, Leadership Workshop, July 2013, Arlington, VA

StartingBloc, Institute for Social Innovation, February 2013, Los Angeles, CA

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International Sociological Association RC28 Conference, Labor Market and Educational Transitions in Uncertain Times, August 2012, Charlottesville, VA

Conference on Asian Pacific American Leadership, Washington Leadership Program, Summer 2012, Washington, DC

Clinton Global Initiative, Clinton Global Initiative University, March 2012, Washington, DC

National Academy of Social Insurance, Social Insurance in a Market Economy: Obstacles and Opportunities, January 2012, Washington, DC

Khan, I. "Develop U: Empowering students, families, and communities through education," October, 2010.

Khan, I. and Springer, K. "Evaluation and implications of drop-out prevention program among minority adolescents," December, 2009.

Khan, I. "The SIOP model: Making content comprehensible for English language learners," July, 2009.

Khan, I. "How MAS Youth fulfills the needs of young people," October, 2008.

Khan, I. "MAS Youth worker orientation and training," July, 2008.

Khan, I. and Ba, Y. "Computational molecular dynamics studies of drug-loaded fluorocarbon-based micelles," May, 2007.

Khan, I. and Ba, Y. "Structure and function of zinc finger proteins," May, 2006.

Khan, I. and McCurdy, A. "Design of a novel transition metal-binding spiropyran catalyst for use in asymmetric synthesis," February, 2005.

Khan, I. and Giedroc, D. "Protein peptide purification of Human Metal-responsive Transcription Factor 1," August, 2003.

Awards and Honors

Rice University Jones School Dean's Scholarship Award, 2020

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Islamic Scholarship Fund Ilhan Omar Courage and Leadership Scholarship, 2018

Islamic Scholarship Fund Deah Yusor Razan Scholarship, 2017

New York City Mayor's Graduate Scholarship, 2017

Kresge Foundation Population Health Scholarship, 2015

New York State Canal Ambassador, 2014

Asian American Government Executives Network Scholarship, 2013

University of Virginia Batten Professional Development Grant, 2011

University of Virginia Batten Merit-Based Fellowship, 2011

Segal AmeriCorps Education Award, 2010

United States Department of State International Visitor Leadership Program Host, 2008

Robert A. Welch Foundation Scholarship, 2003

Texas A&M University Agriculture and Life Sciences Dean's Honor Role, 2001

Texas A&M University Agriculture and Life Sciences Distinguished Student, 2000

Skills and Techniques

Python packages: GeoPandas, Matplotlib, NumPy, Pandas, Plotly, Scikit-learn, Seaborn, StatsModels

Coding and programming languages/software: Amazon AWS Redshift, ArcGIS, AutoCAD, C++, CARTO, Cerius2, CHARMM, Empower, Gaussian, Insight II, LaTeX, MATLAB, Microsoft Power BI, Microsoft SQL Server, PostgreSQL, R, Salesforce, Stata, Tableau, UiPath
Process Mining

Microsoft Office Excel plug-ins: Oracle Crystal Ball, Rutgers YASAI, and Solver

Instrumentation and lab techniques: Column chromatography, Dissolution, Electron microscopy (SEM/TEM), EPR/NMR spectroscopy, HPLC/GC/MS, LIDAR, Materials failure analysis,

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Metallurgy, Organic synthesis, PCR, Polymer design, and Protein purification